ECON345 Homework 4: Code Hygiene, `case_when()`, and R Markdown

Due: September 24th, 11:59 PM

This week we practice writing code a reader can follow — clear names, the right function for the job, and a reproducible **R Markdown** write-up. We'll use the `CPS1985` dataset from the **AER** package throughout, so load `AER` and run `data("CPS1985")` before you start. Use `?CPS1985` to read the documentation.

Comment your code to explain what you're doing — partial credit rewards a reasonable, well-documented attempt, not just a correct final answer.

Part I: Clean Up This Code (8 pts)

The chunk below runs, but it's a mess: cryptic names, no comments, and an "endless sea" of nested `if_else()`. Rewrite it applying today's code-hygiene practices.

```
x <- CPS1985
x$y <- if_else(x$education < 12, "A",
        if_else(x$education == 12, "B",
        if_else(x$education < 16, "C", "D")))
z <- x %>% group_by(y) %>% summarize(m = mean(wage))
```

Your rewrite should:

Question 1 (2 pts)
Open with a **header comment** (2–3 lines) stating what the script does, its input (the data), and its output (the summary table).

Question 2 (2 pts)
Replace the cryptic names (`x`, `y`, `z`, `m`) with **short but descriptive** `snake_case` names.

Question 3 (3 pts)
Replace the nested `if_else()` with a single **`case_when()`** that maps `education` to *meaningful* labels — e.g. `"Less than HS"`, `"High school"`, `"Some college"`, `"College+"`. Make sure your conditions don't overlap and cover every value.

Question 4 (1 pt)
Add a one-line **section comment** above the summary step describing what it computes.

Part II: Conditional Summary Statistics (8 pts)

Now use `case_when()` on a *continuous* variable.

Question 5 (4 pts)

`experience` (years of potential work experience) is continuous, so we can't `group_by()` it directly. Using `mutate()` and `case_when()`, create a categorical variable `experience_level` with **three** groups:

- `"Early career"` — fewer than 10 years
- `"Mid career"` — at least 10 but fewer than 25 years
- `"Late career"` — 25 years or more

Save the result to a new data frame `cps`.

Question 6 (3 pts)

Using `group_by()` and `summarize()`, compute the **mean** wage, **median** wage, and the **number of workers** (`n()`) for each `experience_level`. Save the result to a new data frame.

Question 7 (1 pt)

Interpret. In 1–2 sentences: which experience group earns the most on average, and does the pattern match what you'd expect?

Part III: Assemble an R Markdown Document (4 pts)

Turn Part II into a short, reproducible write-up.

Question 8 (2 pts)

Create a new **R Markdown** file (`week_04.Rmd`), output to **PDF**. Move your Part II code into R code chunks. Set the setup chunk so package-loading messages don't appear in the knitted document (hint: `include = FALSE`).

Question 9 (1 pt)

Present your Part II summary table with **`knitr::kable()`** — round to 2 digits and add a caption.

Question 10 (1 pt)

Below the table, write **one or two sentences of markdown text** (not a code comment) answering Question 7.

Submission

Upload three files to Canvas before the deadline:

- your `.R` script (Parts I–II),
- your `week_04.Rmd`, and
- the **knitted PDF**.

As always: comment your code, and if you get stuck, show your work. If your `.Rmd` won't knit, remember the fixes from lecture — develop the code in a script first, then move it over.